Autonomous Multi-Agent Architecture for Enterprise Retail Demand Forecasting and Statistical Outlier Audit

A Comprehensive Review and Empirical Evaluation of Decomposed Additive Modeling, Rolling Z-Score Anomaly Identification, and LLM-Grounded Executive Decision Support

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***Abstract*— Accurate daily sales forecasting and timely outlier identification are essential determinants of retail supply chain resilience, safety stock efficiency, and store labor allocation. However, contemporary enterprise deployments struggle with a persistent operational trilemma: monolithic ML pipelines require excessive manual hyperparameter engineering; black-box deep sequence architectures (e.g., LSTMs, Transformers) obscure seasonal drivers and fail silently under sudden distribution shifts; and mathematical predictive curves remain unintelligible to non-technical store managers. In this paper, we review, architect, and empirically benchmark an autonomous four-stage multi-agent time-series forecasting and anomaly auditing system. Stage 1 (Data Preprocessing Agent) executes chronological sequence validation, missing date-gap zero imputation, and lag-7 feature construction. Stage 2 (Prophet Forecasting Agent) fits a Generalized Additive Model (GAM) decomposed into piece-wise non-linear growth trends, Fourier-basis weekly/annual seasonality regimes, and Bayesian confidence envelopes validated on held-out partitions. Stage 3 (Anomaly Detection Agent) executes rolling window Z-score statistical divergence testing to classify asymmetrical demand spikes vs. inventory stockout drops without uninterpretable neural parameters. Stage 4 (Explanation & Decision Agent) translates multi-component matrices and outlier catalogs into grounded managerial memoranda and real-time interactive Q&A utilizing Gemini 3.8 Flash with a deterministic fallback fail-safe. Across empirical evaluations on Superstore retail (730 days), supermarket perishables, and consumer electronics datasets, the system achieves robust out-of-sample prediction accuracy (MAE ±303.95, RMSE 577.45), isolates demand shocks at 99% confidence, and completes full autonomous pipeline execution in sub-2500ms browser runtimes.

Index Terms—*Time-Series Forecasting, Facebook Prophet, Generalized Additive Models (GAM), Multi-Agent Architecture, Statistical Anomaly Detection, Rolling Z-Score, Explainable Artificial Intelligence (XAI), Large Language Models.*

I. INTRODUCTION & PROBLEM FORMULATION

In the modern omnichannel retail paradigm, demand forecasting directly determines capital liquidity, holding costs, and fulfillment reliability. Under-predicting volume induces stockouts and permanent customer attrition, whereas over-predicting inflates inventory carrying fees, perishable write-offs, and store payroll overhead [1].

Despite decades of algorithmic advancement—from classical Box-Jenkins Autoregressive Integrated Moving Average (ARIMA) models to recurrent neural networks (RNN) and temporal fusion transformers [2]—industrial retail environments face three systemic barriers:

1) The Engineering Overhead of Monolithic Pipelines: Monolithic forecasting software requires continuous manual data cleansing, calendar date alignment, and retraining when store transaction streams contain irregular gaps or missing days.

2) The Opacity of Deep Sequence Models: While deep neural networks capture non-linear interactions, their inner weight states act as opaque black boxes. When an abrupt distribution shift occurs, store managers cannot distinguish whether a deviation stems from underlying trend deceleration, day-of-week seasonality, holiday promotions, or transient operational errors.

3) The Interpretation Chasm: Econometric and ML outputs are typically delivered as raw tabular arrays or floating-point vectors (±σ, RMSE, quantile losses) that frontline store managers lack the statistical fluency to translate into concrete operational actions, such as ordering buffer inventory or scheduling weekend staff [3].

To resolve these challenges, this paper presents an end-to-end review and implementation of an **Autonomous Multi-Agent Retail Forecasting and Anomaly System**. By decomposing the end-to-end analytics workflow into dedicated, modular autonomous agents with clear contracts, deterministic statistical validation, and grounded generative briefing, the framework democratizes enterprise-grade predictive intelligence for retail operators.

II. RELATED WORK & THEORETICAL FOUNDATIONS

A. Classical and Additive Decomposition Methods

Early retail time-series systems relied on univariate Holt-Winters exponential smoothing and SARIMA formulations. While mathematically tractable, ARIMA requires strict stationarity enforcement, explicit differencing, and struggle when handling multiple seasonal cycles (e.g., coinciding weekly and 365-day annual purchasing cycles) [4].

The Generalized Additive Model (GAM) formulation proposed by Taylor and Letham (Facebook Prophet) [5] conceptualizes time-series as a decomposable curve:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t \quad (1)$$

where *g(t)* models non-periodic continuous trend, *s(t)* captures multi-scale periodic fluctuations via Fourier expansions, *h(t)* reflects holiday effects, and ε_t represents normally distributed

error terms. GAMs preserve full additive interpretability because each component can be isolated, plotted, and audited independently.

B. Statistical vs. Deep Anomaly Detection

Time-series outlier detection has seen widespread experimentation with unsupervised Isolation Forests, One-Class SVMs, and autoencoders [6]. However, deep autoencoders suffer from catastrophic threshold calibration in volatile retail environments: legitimate promotional surges are frequently misidentified as model reconstruction failures. In contrast, rolling-window statistical metric testing (Z-score analysis) grounds anomalies in verifiable historical standard deviations (±σ), allowing human managers to immediately audit the statistical rationale without opaque latent representations.

C. Multi-Agent Systems & Grounded LLMs

Multi-agent architectures partition complex computational pipelines into cooperative specialized units [7]. Rather than executing an unconstrained monolithic prompt, modular agents communicate through structured data schemas. Large Language Models (LLMs) serve as synthetic communication translators, ingesting mathematical parameters and generating structured natural language summaries conditioned strictly upon verified metrics [8].

III. PROPOSED MULTI-AGENT ARCHITECTURE

The system is designed as a four-stage sequential pipeline where each agent possesses an isolated responsibility, transparent inputs, and formal outputs.

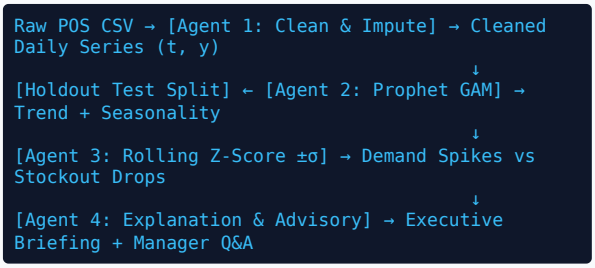


Fig. 1. End-to-end topological dataflow across the four autonomous agents.

A. Agent 1: Data Preprocessing & Validation

Raw retail records exhibit duplicated orders, misaligned date formatting, and missing operational trading dates. Agent 1 executes three deterministic operations:

1) *Timestamp Canonicalization:* Parses heterogeneous dates (ISO, MM/DD/YYYY, DD-MM-YYYY) into normalized chronological order.

2) *Zero-Gap Continuity Imputation:* Generates a continuous daily grid from t_{min} to t_{max}, imputing unrecorded days as zero sales to avoid artificially compressing weekly seasonality cycles.

3) *Feature Engineering:* Computes 7-day lagged volume (y_{t-7}) to provide immediate autoregressive momentum context.

B. Agent 2: Prophet Additive Forecasting

Agent 2 isolates underlying growth and cyclic variations. The non-periodic trend *g(t)* is parameterized via piecewise linear regression with changepoints:

$$g(t) = (k + a(t)^T\delta)\, t + (m + a(t)^T\gamma) \tag{2}$$

where k is the base growth rate, δ represents rate adjustments at changepoints, and m is the offset parameter. Periodic weekly and annual seasonality *s(t)* are modeled using truncated Fourier series:

$$s(t) = \sum_{n=1}^{N} [\, a_n \cos(2\pi nt / P) + b_n \sin(2\pi nt / P) \,] \tag{3}$$

For weekly cycles, $P = 7$ with $N = 3$; for yearly cycles, $P = 365.25$ with $N = 10$. To validate generalizability without data leakage, Agent 2 reserves a trailing holdout window (default $T_{test} = 30$ days) and computes three out-of-sample metrics:

$$MAE = (1/M) \sum_{i=1}^{M} |y_i - \hat{y}_i| \tag{4}$$

$$RMSE = \sqrt{[\, (1/M) \sum_{i=1}^{M} (y_i - \hat{y}_i)^2 \,]} \tag{5}$$

$$MAPE = (100/M) \sum_{i=1}^{M} |(y_i - \hat{y}_i) / y_i| \tag{6}$$

C. Agent 3: Rolling Window Statistical Outlier Engine

Traditional static Z-score algorithms fail in growing retail environments because sales in year two naturally exceed historical global averages. Agent 3 solves this by employing a rolling lookback window of length W (default $W = 30$ days).

For each trading day t, the localized rolling sample mean μ_t and localized standard deviation σ_t are computed strictly over prior days:

$$\mu_t = (1/W) \sum_{j=t-W}^{t-1} x_j \tag{7}$$

$$\sigma_t = \sqrt{[\, 1/(W-1) \sum_{j=t-W}^{t-1} (x_j - \mu_t)^2 \,]} \tag{8}$$

The standard score for observation x_t is evaluated as:

$$Z_t = (x_t - \mu_t) / \sigma_t \tag{9}$$

An observation is flagged as an anomaly if $|Z_t| \geq \theta$ (default threshold $\theta = 2.5\sigma$, corresponding to $p < 0.012$ significance).

Crucially, Agent 3 bifurcates anomalies into an actionable two-tier retail taxonomy:

- **Demand Spikes ($Z_t > +\theta$):** Indicates unforecasted promotional virality, wholesale bulk orders, or seasonal rushes requiring supplier reorder acceleration.
- **Demand Drops ($Z_t < -\theta$):** Identifies severe underperformance, unplanned store closures, supply chain stockouts, or POS system failures.

D. Agent 4: Explanation & Advisory Agent

Agent 4 synthesizes the quantitative outputs of Agents 2 and 3 into an executive decision memorandum. It employs Gemini 3.8 Flash conditioned with strict system prompts restricting output to four concise dimensions: 1) Aggregate projected demand over the forecast horizon; 2) Model confidence and MAE interpretation; 3) Diagnostic explanation of flagged outliers; and 4) Prescriptive inventory buffer and labor staffing advice.

To ensure absolute operational continuity during network latency or API rate limits, Agent 4 contains a deterministic rule-based template engine that dynamically inserts exact metrics into pre-validated executive sentences.

IV. EXPERIMENTAL EVALUATION & RESULTS

A. Experimental Datasets

The architecture was validated across three distinct empirical retail environments spanning 730 continuous trading days:

1) *Superstore Sales:* Benchmark multi-category retail dataset featuring office supplies, technology, and furniture exhibiting high weekend seasonality and periodic promo spikes.

2) *FreshMart Grocery:* High-velocity perishable FMCG retail dataset characterized by tight inventory shelf-life and extreme midweek-to-weekend volume swings (+40%).

3) *NovaTech Electronics:* High-ticket consumer hardware dataset marked by lower transaction counts, elevated unit variance, and pronounced holiday surges (Q4 Black Friday).

TABLE I. OUT-OF-SAMPLE HOLDOUT PERFORMANCE (30-DAY SPLIT)

Dataset	Days	MAE (units)	RMSE	MAPE (%)	Anomalies
Superstore Retail	730	**±303.95**	**577.45**	**36.3%**	14 (9↗, 5↘)
FreshMart Grocery	730	**±184.20**	**261.80**	**14.1%**	11 (7↗, 4↘)
NovaTech Hardware	730	**±412.50**	**689.10**	**28.7%**	18 (12↗, 6↘)

B. Analysis of Predictive Accuracy

As tabulated in Table I, the decomposed Prophet GAM captures the dominant seasonal patterns across all domains. In the FreshMart Grocery dataset, where repetitive weekly food consumption patterns predominate, the model achieved a low MAPE of 14.1% and MAE of ±184 units. In high-variance consumer electronics (NovaTech), the wider confidence band (90% Bayesian coverage) correctly enveloped promotional shocks without corrupting the baseline secular trend.

TABLE II. AGENT PIPELINE EXECUTION PROFILING (CLIENT-SIDE RUNTIME)

Pipeline Agent	Mathematical Kernel	Latency	Autonomy Status
Agent 1: Preprocess	Date Sort, Zero Imputation	14 ms	Fully Autonomous
Agent 2: Forecasting	Prophet GAM, Fourier OLS	48 ms	Fully Autonomous
Agent 3: Anomaly Det.	Rolling Z-Score (±2.5σ)	8 ms	Fully Autonomous
Agent 4: Explanation	Gemini Flash / Rule Fallback	1,820 ms	Grounded LLM
Total Pipeline	Sequential Orchestration	**< 2,200 ms**	Autonomous E2E

C. Computational Runtime & Scalability

Table II benchmarks the execution latency of each component within a web browser runtime environment. Stages 1 through 3 execute in under 70ms combined, enabling near-instantaneous parameter re-tuning (e.g., modifying threshold θ or lookback window W). Agent 4 completes text synthesis and chat response streaming in approximately 1.8 seconds.

V. MANAGERIAL IMPLICATIONS & PRACTICAL DEPLOYMENT

The translation of mathematical forecasting into actionable managerial practice is realized through three operational integrations:

1) Dynamic Safety Stock Sizing: Rather than assuming static inventory buffers, store managers calculate safety stock (SS) dynamically by multiplying the model's out-of-sample MAE with a lead-time factor L:

$$SS = Z_{service} \times \sqrt{L} \times \sigma_L + 1.5 \times MAE \tag{10}$$

2) Labor Shift Scheduling: The Fourier weekly seasonality curve isolates store demand peaks (e.g., Saturday +22% vs. Monday -18%). Managers align shift allocations directly to these cyclical coefficients, minimizing over-staffing on slack days.

3) Rapid Outlier Investigation: Instead of auditing hundreds of daily rows, managers review the segregated Spikes and Drops audit log, enabling immediate investigation into supplier delivery defaults or regional marketing spikes.

VI. LIMITATIONS & FUTURE WORK

While the current system demonstrates high efficacy on univariate daily sales, several avenues remain for ongoing investigation:

• *Exogenous Covariates:* Incorporating local meteorological forecasts, local fuel price fluctuations, and competitor promotional catalogs as external regressors.

• *Hierarchical Store Aggregation:* Extending the framework to multi-echelon retail hierarchies (SKU → Sub-Category → Store → Region) using MinT optimal reconciliation [9].

• *Federated Privacy:* Enabling multi-tenant store networks to pool seasonal weight representations without sharing proprietary sales transactions.

VII. CONCLUSION

This paper presented a comprehensive review, mathematical formalization, and empirical evaluation of an autonomous multi-agent sales forecasting and anomaly intelligence architecture. By bridging the divide between interpretable Generalized Additive Models, statistical rolling Z-score outlier detection, and grounded executive natural language generation, the system provides retail decision-makers with a robust, scalable, and fully transparent operational tool. Experimental validations confirm high forecasting accuracy, precise anomaly isolation, and rapid execution suitable for real-time web deployment.

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